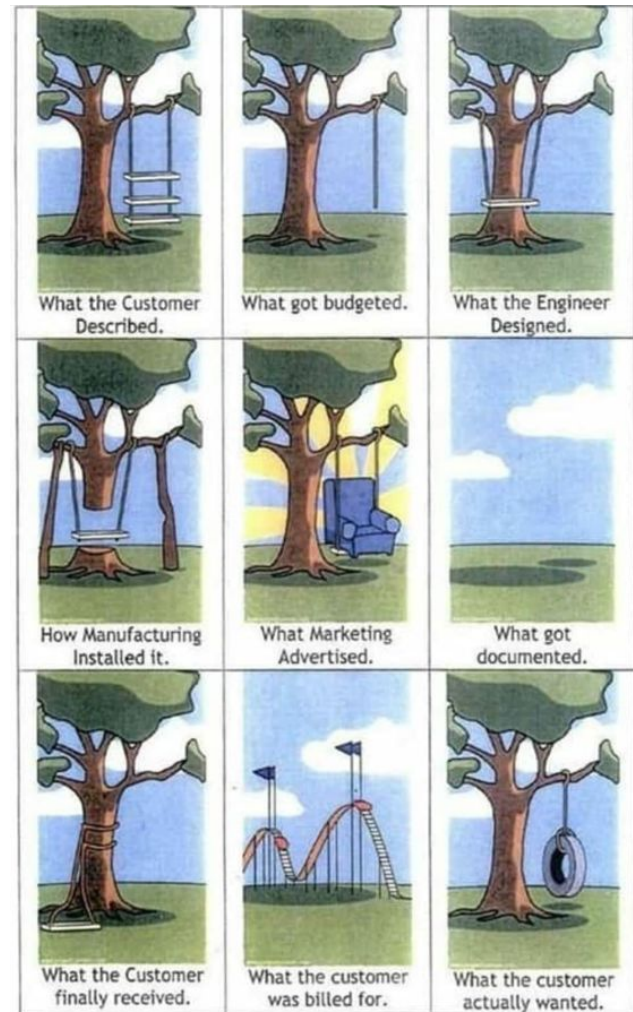


Design Brief

The foundational document that guides the project



Welcome to Your Design Brief!

Purpose: Kick off your Week 2 **project** to design a desktop cell phone stand using the Engineering Design Process (EDP).

Overview:

- This week, you'll create a design brief for a item you will design.
- Focus: Ask, Research, Imagine steps, evaluated with a Pugh Matrix (Plan step).
- Work in teams, use Chromebooks for research, paper/pencil for sketches. Mr. Bombich is your "customer."
- Due: Monday, Sept 8, 2025, noon (Google Classroom, Google Doc with sketches/photos).

What To Expect

- **This Week:**
- (Tue): Problem Statement (Ask).
- (Wed): Research Summary (Research).
- (Thu): Brainstormed Ideas (Imagine).
- (Fri): Pugh Matrix (Plan) & draft finalization.

Tip: Check the rubric on Google Classroom for expectations! Keep journals to document your role.

Problem Statement (Ask Step)

Purpose: Define the problem clearly to start the EDP

Instructions:

- Describe the problem: Who is affected? What's the issue? Where/when does it occur? Why does it matter? How big is it?
- List stakeholders (e.g., users, organizations).
- Specify needs (e.g., functionality, ease of use).
- Detail constraints: budget, manufacturing capabilities, materials, time, size.
- Include ethical considerations (e.g., sustainability, safety).

Problem Statement (Ask Step)

Rubric Criteria (80–89%): Clear problem with stakeholders, needs, constraints, and ethics. Relevant questions for the customer.

Example: “Manual recycling sorting is slow and error-prone. Stakeholders: school staff, environmental groups. Needs: accurate sorting. Constraints: \$50 budget, Arduino sensors. Ethics: low energy use.”

Tip: Be specific! E.g., “sort 80% of items correctly” vs. “sort well.”



Research Summary (Research Step)

Purpose: Gather info to inform your solution.

Instructions:

- Use Chromebooks to find 3–5 credible sources (prompts: DIY cell phone stand, Amazon cell phone stand, etc.).
- Summarize findings on existing solutions or components (e.g., folding cell phone stands, articulating cell phone stands).
- Connect research to project constraints (e.g., cost) and ethics (e.g., energy efficiency).
- Cite sources fully (source, date; include AI use, e.g., “Grok by xAI, Sept 3, 2025”).

Research Summary (Research Step)

Rubric Criteria (80–89%): Summarize 3–5 sources with relevant insights tied to constraints/ethics. Citations mostly complete.

Example: “Instructables.com shows IR sensors (\$2–5) for material detection. EPA.gov says 25% of recycling is contaminated. Low-power sensors support sustainability.”

Tip: Use Google Docs to organize sources and upload to Google Classroom daily.

Brainstormed Ideas (Imagine Step)

Purpose: Generate creative solutions with your team

Instructions:

- Brainstorm 5+ ideas in a team mind-mapping session.
- For each idea: Write 2–3 sentences describing it, list pros/cons, and note how it meets constraints/ethics.
- Create paper/pencil sketches for each idea (photo-upload to Google Docs). Label clearly.
- Document team roles in journals (e.g., who sketched, brainstormed).

Brainstormed Ideas (Imagine Step)

Rubric Criteria (80–89%): 5 ideas with clear descriptions, adequate sketches, addressing constraints/ethics. Collaboration evident in journals.

Example: “Idea: Ramp sorter. Items slide down, flaps divert to bins. Pros: Simple. Cons: Less accurate. Sketch: Ramp with flaps, bins.”

Tip: Go wild with ideas first, then check constraints! Clear sketches help.

Evaluation (Pugh Matrix)

Purpose: Choose the best idea using a decision matrix

Instructions:

- Create a Pugh Matrix table comparing all ideas to a baseline (e.g., Idea 1).
- Use 3–5 criteria (e.g., functionality, cost, ethics) with weights (1–5, higher = more important).
- Score ideas: + (better, +1), S (same, 0), - (worse, -1). Multiply score by weight, sum for total.
- Justify the top-scoring idea, tying to project goals and ethics.

Evaluation (Pugh Matrix)

Rubric Criteria (80–89%): Complete matrix with 3–5 criteria, mostly accurate math, clear justification with some ethical focus.

Example: “Criteria: Functionality (5), Cost (3). Idea 3: $-1 \times 5 + 1 \times 3 = -2$. Pick Idea 3: simple, sustainable.”

Tip: Double-check math! Use a template from Google Classroom.

Evaluation (Pugh Matrix)

Criteria	Option	Option	Option
Cost	–	+	0
Durability	0	0	–
Ease of Fabrication	+	+	–

Pugh Matrix Alternative Solutions

Selecting a new solution for online file storage



CLOUDER



RainbowCloud



CLOUDATUM



MINDPLACE STORAGE



Criteria	Current Solution	Alternative 1	Alternative 2	Alternative 3
Ease of use	0	-1	+1	+1
Storage limits	0	+1	-1	+1
Cost per seat	0	-1	-1	-1
Customer Support	0	+1	-1	+1
Time to transfer existing files	0	+1	0	+1
Total Score		+1	0	+3

Notes: Based on the results of the Pugh Matrix, MindPalaceStorage offers several advantages compared to Clouder, so making the switch is recommended.



GLOBAL
BUSINESS
SOLUTION

Submission & Next Steps

Submission Details:

- Submit as a Google Doc via Google Classroom by Monday, Sept 8, 2025, noon.
- Include sections: Problem Statement, Research Summary, Brainstormed Ideas, Pugh Matrix, Next Steps, References.
- Upload photos of paper/pencil sketches (clear, labeled).
- Late penalty: -10% per day (max 3 days, per syllabus).

Tip: Review the rubric before submitting. Ask Mr. Bombich for feedback as the “customer” on Day 8!

Spot the Difference

Section	Weak Brief	Strong Brief
Context	"Cables are messy."	"Our lab benches have three loose cables—power, USB, network—that create trip hazards and slow setup."
Problem Statement	"Make a cable clip."	"Design a 3-cable organizer that clips to a 1–1.5" tabletop so users can quickly attach/detach without tangles."
Requirements	"Must hold cables."	"Must hold up to 3 cables (≤ 6 mm $\varnothing$), allow insertion in ≤ 2 lb force, removal in ≤ 1 lb force."
Constraints	"Use plastic."	"Use PLA filament; cost $\leq \$1$ per unit; fit in 120×120×120 mm print volume; complete by Week 3."
Success Criteria	"Works well."	"90 % of 10 test users attach cable in ≤ 5 s; clip endures 100 cycles without damage."

Cable Management Challenge

Objectives

- Translate sketches into a functional prototype
- Practice rapid ideation and low-fidelity fabrication
- Evaluate usability with real cables



Cable Management Challenge



Cable Management Challenge

Materials

- Sketch paper & pens
- Cardboard scraps, modeling clay, tape
- 3 types of cables (USB, power, Ethernet)
- Stopwatch (optional)

Cable Management Challenge

Procedure

1. **Rapid Sketch (5 min)**
 - Draw **3** distinct cable-clip concepts
2. **Peer Critique (5 min)**
 - Each team shares one sketch; feedback using “I like..., I wonder...”
3. **Build Prototype (15 min)**
 - Use cardboard/clay/tape to construct your best concept
4. **Usability Test (10 min)**
 - Install/remove each cable 3×; record ease and issues

Cable Management Challenge

Test Data Table

Clip Concept	Install Time (s)	Remove Time (s)	Notes (fit, stability)
Concept 1			
Concept 2			
Concept 3			

Cable Management Challenge

Team Name: _____

Date: _____

1. What Worked Well?

2. Biggest Usability Issue

4. One Improvement for Next Time

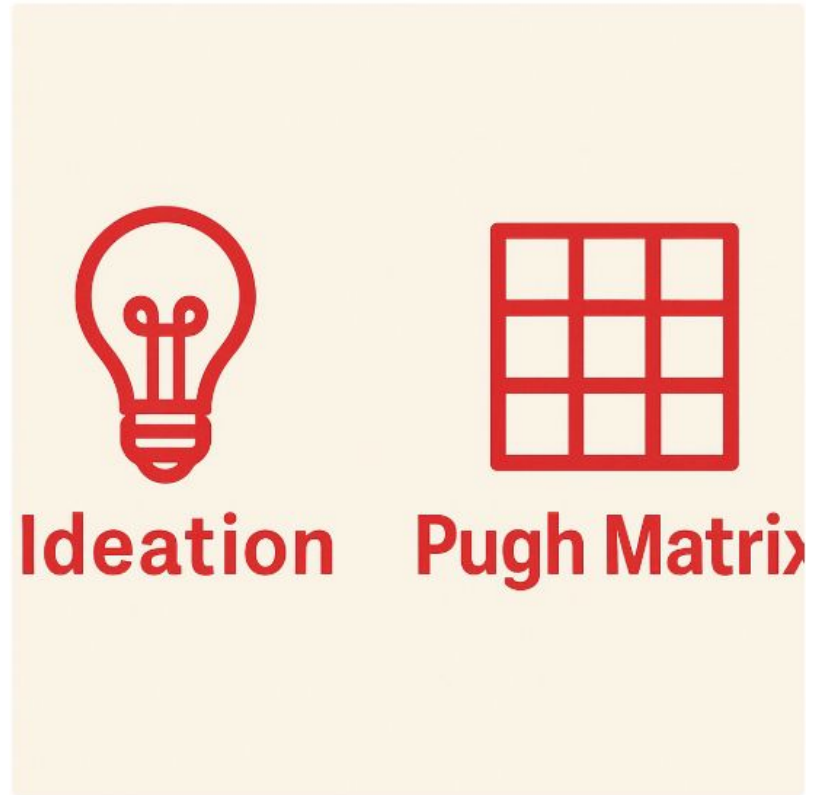
Ideation Deep-Dive

Objectives:

- Refine and converge on your top two concepts
- Evaluate ideas against project criteria using a Pugh Matrix
- Capture rationale for concept selection

Materials:

- Your Week 1 design brief and initial sketches
- Sketch paper & pens
- Pugh Matrix worksheet (6×6 grid)
- Sticky dots or markers for scoring

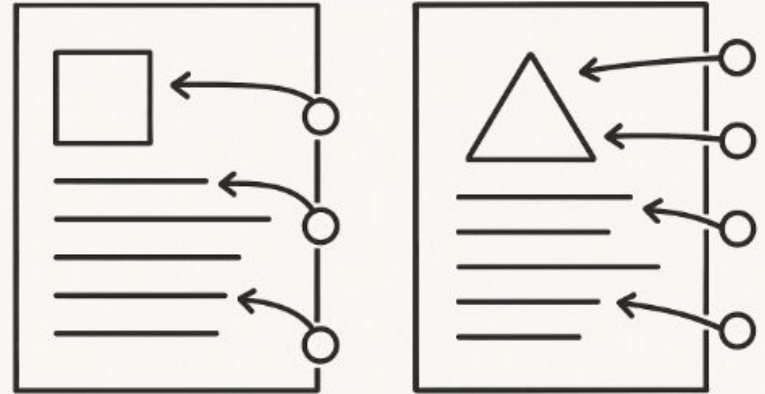


Rapid Sketch (10 min)

- **Select Two Concepts:** Choose your top two initial ideas.
- **Refine Details:** Sketch each at a larger scale, adding key features.
- **Annotate:** Note how each meets requirements & constraints.

Prompt:

“Which design elements will most impact usability and feasibility?”



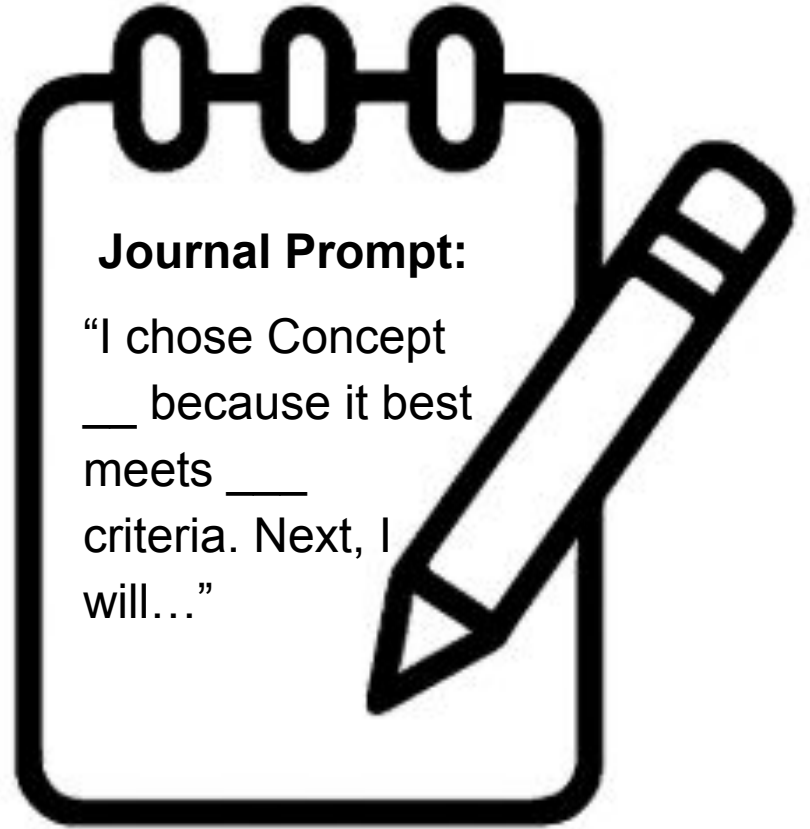
Reflection

Discussion:

“Which concept scored highest—and why?”

Journal Prompt:

“I chose Concept ___ because it best meets ___ criteria. Next, I will...”



EXIT

**Write one
question you
have about
moving this
concept into
CAD.**